Name: ___________________________________

Year 12 Essentials

Mathematical thinking process questions

Baldivis
Secondary College

You will need to apply the mathematical thinking process:

- interpret the task and gather the key information
- identify the mathematics which could help to complete the task
- analyse information and data from a variety of sources
- apply their existing mathematical knowledge and strategies to obtain a solution
- verify the reasonableness of the solution
- communicate findings in a systematic and concise manner.

Scenario:

The Baldivis Secondary College's Student Representative Council (SRC) has a long term goal to raise money to purchase and install a large covered veranda. This veranda will provide students with additional shelter in hot or wet weather.

Task 1: Investing funds

The SRC has already raised $6000 from previous fund raising activities and wishes to determine the best way to invest this money while additional funds are raised. In this task you are required to compare four different investment options. The three different investment options are listed in the table below.

OPTION 1	6% per annum	No fees charged
OPTION 2	7% per annum	$50 annual fee
OPTION 3	3.2% per half year	No fees charged

Q1. For each option you need to:
   a. calculate the interest earned over a 5 year period;
   b. add this to the original amount that was invested;
   c. Deduct any fees.

Q2. Which of the three options is best? Justify your answer.

Q3. Using Option 1, how many years would it take to raise $1000 in interest?

Q4. Suppose you wished to raise $2000 interest in 10 years. What interest rate would be required? Is this interest rate reasonable?

1. $\$6000 \times 0.06 \times 5$
$= \$1800$ interest after 5 years
$\$6000 + 1800$
$= \$7800$ end amt
no fees

2. $\$6000 \times 0.07 \times 5$
$= \$2100$ interest
$\$6000 + \$2100$
$= \$8100$
$8100 - (\$50 \times 5)$
$= \$7850$ end amt

3. $\$6000 \times 0.032 \times 10$
$= \$1920$ interest
$\$6000 + \$1920$
$= \$7920$ end amt
no fees

Task 2: Borrowing money

The SRC has decided that they do not want to wait 5 years to purchase and build a shelter. They have determined that it will cost $15000 to build a shelter. The SRC will use the $6000 they already have and borrow the remaining amount. They have received three options for loans, all based on simple interest, as shown in the following table.

OPTION 1	8% per annum	To be paid back over 10 years
OPTION 2	7% per annum	To be paid back over 12 years
OPTION 3	0.5% per month	To be paid back over 15 years

Q5. For each option calculate:

a. The total interest paid
b. The total amount to be repaid
c. The amount paid per year.

Q6. Which option is best? Justify your answer.

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Task 2: Compound Interest

In the investigation to date, all calculations have used simple interest. In reality financial institutions mostly use compound interest in their calculations.
To calculate compound interest, the amount of interest earned in the previous is added to the principal to determine the amount of interest earned in the current year. So in effect interest is earned on interest.

Q5. Given an initial investment of $6000 that is paying 6% compound interest per annum calculate how much the investment will be worth in total after:
   a. 1 year
   b. 2 years
   c. 3 years
   d. 4 years
   e. 5 years

Q6. How does this compare with investment Option 1 from Task 1? Which option would you go for? Why are the two options different?

Compound interest can also be calculated using the following formula:

$$FV = PV(1+i)^n$$

Where FV is the future value of the investment, PV is the present value (or starting value) of the investment, i is the interest rate (as a decimal), and n is the number of years.

Q7. Use the above formula to calculate the value of an initial investment of $6000 paying 6% compound interest per annum at the end of 5 years.

Q8. Compare your answer to Q7. Is the answer the same? Which method is easier to apply? Would your answer change if you had to calculate the interest over 30 years?

$i = p \times r \times t$

	initial amt	simple interest	new amt.
yr 1	$6000	$6000 \times 0{\cdot}06 \times 1 = 360$	$6000 + 360$ $\boxed{= \$6360}$
yr 2	$6350	$6350 \times 0{\cdot}06 \times 1$ $= \boxed{381{\cdot}6}$	$6360 + 381{\cdot}6$ $\boxed{= \$6741{\cdot}6}$
yr 3	$6741·6	$6741{\cdot}6 \times 0{\cdot}06 \times 1$ $= \boxed{404{\cdot}5}$	$6741{\cdot}6 + 404{\cdot}5$ $\boxed{= \$7146{\cdot}1}$
yr 4	$7146·1	$7146{\cdot}1 \times 0{\cdot}06 \times 1$ $= \boxed{428{\cdot}77}$	$7146{\cdot}1 + 428{\cdot}77$ $\boxed{\$7574{\cdot}87}$
yr 5	$7574·87	$7574{\cdot}87 \times 0{\cdot}06 \times 1$ $= \boxed{454{\cdot}49}$	$7574{\cdot}87 + 454{\cdot}99$ $\boxed{= \$8029{\cdot}36}$

Q6. option 1 task 1
simple interest
= $1800 after 5 yrs
total end amount
= $7800

$$6000\left(1 + \frac{0{\cdot}06}{1}\right)^{1 \times 5}$$
$$= \$8029{\cdot}35$$

Answer/Findings/ Solution:

$6000 5% pla 5 years

Q6. option 1 task 1.

simple interest
= $1800 after 5 years

total end amount
= $7800

compound interest
= $8029.35 end amt.

$8029.35 − $7800
= $229.35
difference.

by applying compound interest to the principle of $6000 at a 6% per annum rate for 5 years, after the 5 years the interest is $2029.35, with an end amount equally $8029.35. If simple interest is applied instead, with the same principle of $6000 and 6% per annum rate, after the 5 years the interest earned is $1800 bringing the total to $7800. This is a difference of $229.35 between compound and simple interest in compound interest favour. This is the better option because after the 5 years you earn a further $229.35.

Q7.

$$\$6000\left(1+\frac{0.06}{1}\right)^{1\times 5}$$
$$= \$8029.35$$

Q8. Compare Q5 to Q7.

By using the compound interest table method that looks like the following:

amt	simple interest	new amt

the total amount at the end of the 5 years is $8029.36. By using the compound interest formula of, $P(1+\frac{r}{n})^{nxt}$, the total amount at the end of 5 years is $8029.35. There is only a 1¢ difference between the two total and likely due to rounding errors whilst calculating the compound interest through the table method.

The easier of the two methods is the formula due to their being only one calculation instead of five seperate ones for the table method. This would always be the case even if the years increase to upwards of 30. If anything, it actually becomes easier this way.

Mathematical Thinking Process Planning Sheet

Interpret the task	
Identify the mathematics involved	
Analyse information/data (you may need to use other sources)	
Apply existing mathematical knowledge and strategies	
Verify the reasonableness of your answer	
Communicate findings	See back page